MAKERVERSE

LEARN CREATE INNOVATE

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DIPOLOG ROBOTICS FESTIVAL

AND COMPETITION

2026

GAME RULES

Line Dash

"Stay on Track. Dash to Victory"

Objective: The robot must dash along a straight or curvy line track as fast as possible and reach the finish line first.

Reference: World Robot Games Philippines. (2026). *World Robot Games 2026-PHL Online GameRules and Maps* [PDF]. Summer Robot games 2025 https://drive.google.com/drive/folders/1c5R7VDAZjqRO57JxGOiHQyiGtIo910ZG

Section 1: Participant Eligibility

1.1 Each team consists of **one (1)** participant only.

1.2 **Competition Categories:**

- Grades 1-3
- Grade 4-8
- Grade 9-12
- Open Division: No grade restriction (including teachers, mentors, and hobbyists)

1.3 Each participant may compete in only one team and one category.

Section 2: Equipment & Preparation

2.1 Participants must bring:

- Fully assembled and functional robot
- Laptop with programming software
- Batteries or power supply
- One (1) screwdriver for adjustments*(to be used only before the game starts and when signaled by the referee)*

2.2 Inspection

- Robots shall undergo inspection prior to official runs to verify compliance with dimensional, weight, and competition regulations.
- Robots must fit within a box and **must not exceed 24 cm × 24 cm** for inspection(all categories).

Section 3: Playing Field Specifications

Notes: *The information below shows each category along with its designated track layout. Categories are divided by grade levels: Grades 1–3, Grades 4–8, Grades 9–12, and Open Division.*

3.1 Category: Grade 1-3

- **Field dimensions**: 8’ × 12’ , flat and stable
- **Surface**: Printed PVC, vinyl, or equivalent material
- **Line width**: 30 mm black line
- The line starts and ends at designated points
- It may include straight sections, curves, and intersections.

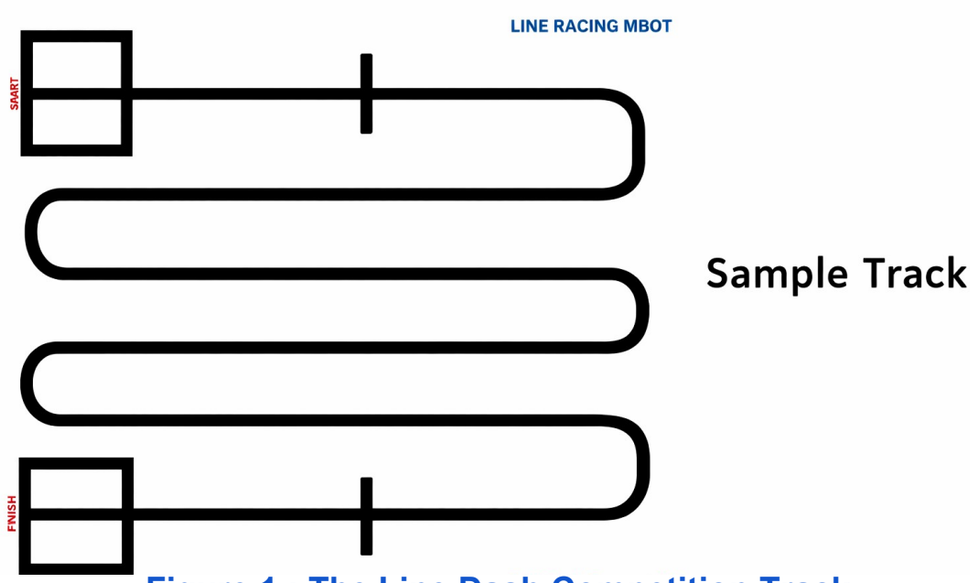
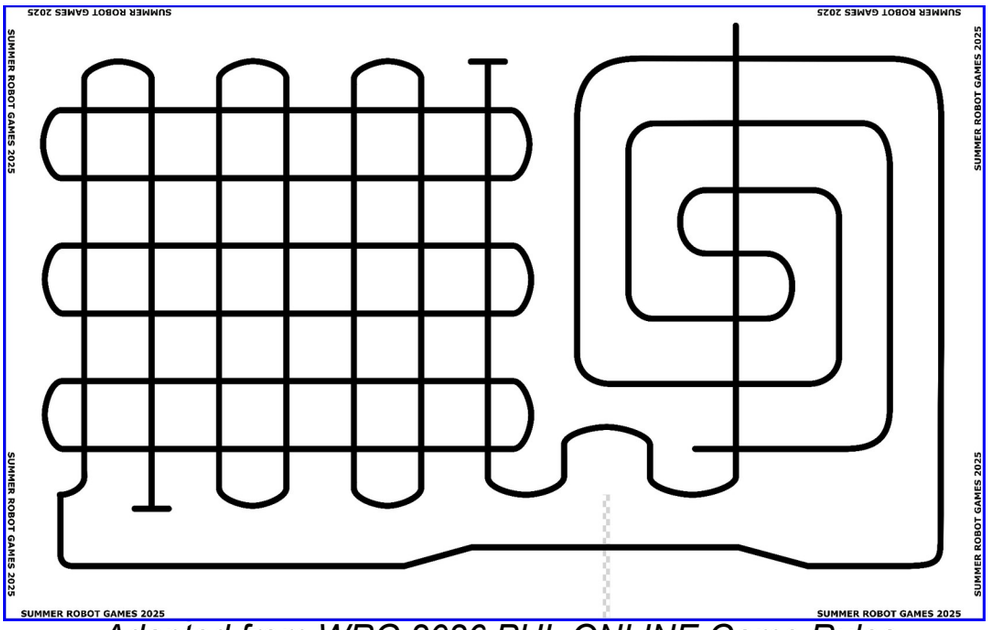


Figure 1 : The Line Dash Competition Track

Adapted from PILIPINAS ROBOSPORTS TOURNAMENT version 8

3.2 Category: Grade 4-8 and Grade 9-12



Adapted from WRG 2026 PHL ONLINE Game Rules

3.3 Category: Open Division

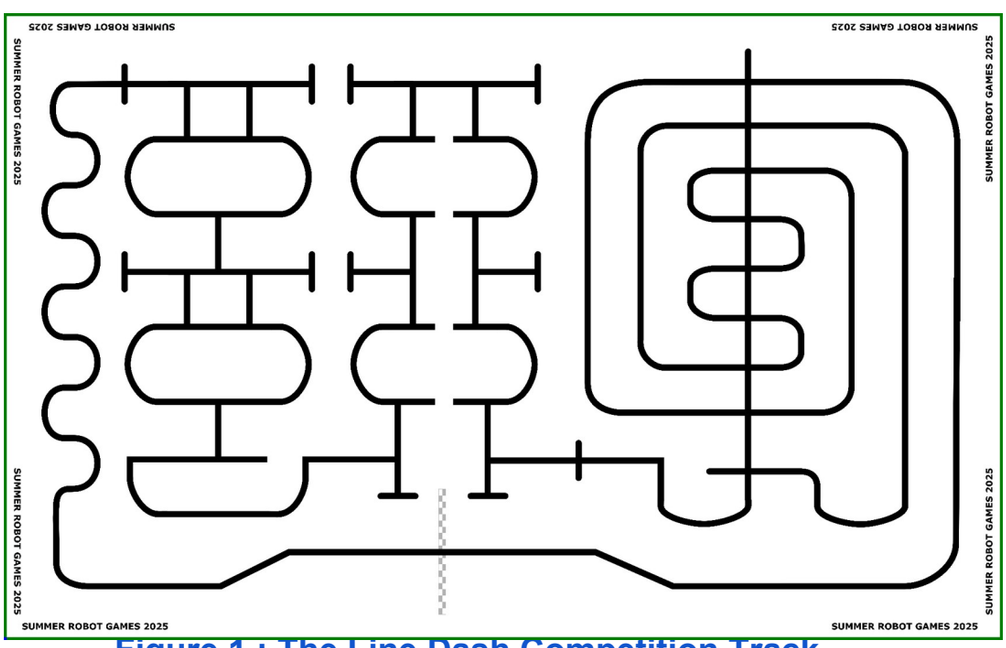


Figure 1 : The Line Dash Competition Track

Adapted from WRG 2026 PHL ONLINE Game Rules

Section 4: Robot Requirements

4.1 Robots must fit within a box and **must not exceed 24 cm × 24 cm**
4.2 No brushless motors
4.3 Must move fully autonomously
4.4 No weight and height limit
4.5 All parts must be securely attached; fallen parts may not be touched until run completion
4.6 Cannot extend size during operation

Section 5: Controller & Components

5.1 Any microcontroller allowed
5.2 Motor & Sensor Limits

- Max. 2 motors
- Max. 4 sensors

5.3 Power & battery: No limitation
5.4 Any color

Section 6: Prohibited Equipment
6.1 Any device that may damage the playing field
6.2 Sticky substances, liquids, powders, or gases
6.3 Parts intended to damage the opponent or the field

Section 7: Competition Format

7.1 Attempts

- Each participant has **two (2) official runs**.
- Extra runs may be allowed with **judge's approval**
- Each participant may take two (2) attempts
- **The fastest valid run** holds the “Best Time.”

7.2 Ranking

- Based on the **fastest valid run, including penalties**.
- **Ties** are broken using other round results.
- **Winners** are declared per division/category based on the best valid run.

*Notes: If **no** robot crosses the finish line, the robot that has traveled the **FARTHEST** will win. If two or more robots reach the same distance, the one with the **SHORTEST TIME** will be ranked highest.*

Section 8: Practice & Robot Calibration

- Makerverse officials will determine the official times during which participants may calibrate and practice their robots on the competition track.
- Participants may practice on the official track only while lining up and carrying their robots.
- Participants are not allowed to step onto the playing track.

Section 10: Game Procedure & Mission

10.1 **Start**

- The robot begins at the **starting point** when signaled.
- Only the participant may start the robot.

10.2 **During the Run**

- The robot must **follow the line**.
- Participants **cannot touch the robot** during the run.
- **Penalties** apply for leaving the track or breaking rules.

10.3 **Scoring**

- **Run time** is recorded from start to finish.
- **Penalties** are added to the run time.
- **Fastest valid run** determines ranking.

10.4 **End of Run**

- The run ends when the robot **crosses the finish line**.
- The referee records **final time and penalties**.
- The robot is **removed from the track** immediately.

10.5 **Touch and Recovery**

- Return the robot to where it left the track or as instructed by the judges.
- Failure to do so ends the run immediately.
- **Time continues** during recovery.
- Each touch adds a **5-second penalty**.

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Section 11: Fouls & Conduct

- **No touching** the robot during a run, except as allowed before the start or during recovery.
- **No interference** with other participants' robots.
- **No modifications** to the robot or program during official runs unless permitted.
- **Disrespectful behavior** toward judges, participants, or staff is prohibited.
- **Violations** may result in **penalties or disqualification**.

Section 12: Robot Identification

12.1 Each robot must display:

- **Team name**
- **Participant name or ID**

12.2 Must be clearly visible to judges

Section 13: Judge Authority

13.1 Judges control:

- Run validity
- Compliance with rules
- Timing and scoring

13.2 Decision hierarchy:

1. Head Judge
2. Panel of judges

Section 14: Final Authority

Note: *The referees are responsible for determining violations and enforcing penalties in accordance with the competition rules. All decisions made by the referees shall be final and binding.*

Robo Push

"Push Beyond Limits"

The *objective* of Robo Push is to **push the opponent robot outside the elevated arena** within **three (3) minutes**. Robots must comply with maximum mass limits per division.

Reference: Pilipinas RoboSports Tournament. (2026, February 5). *Pilipinas RoboSports Tournament version 8 rulebook* (Version 8) [PDF]. https://drive.google.com/drive/folders/1SwscBqoWkHreAloQ6HFQ4x7IYCx-Endd

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Section 1: Participants

Elementary Division (RC)

1.1 Teams: 1–2 members, currently enrolled in elementary school.
1.2 Only **one coach** allowed per team
1.3 **Competition Categories**:

- Grade 4-8
- Grade 9-12

Open Division (Auto & RC):

- Teams: 1–2 members, ages **8–65 years old**
- Only **one coach** allowed per team

Section 2: Robot Requirements

Autonomous Robots

- Any microcontroller allowed
- Maximum dimensions:
  - 3kg: **20cm × 20cm**
  - No height limit
- Must be activated with a **manual switch** (button, toggle, or pull-up switch)

- **Sensors or RFID tags** cannot trigger robot movement

Remote-Controlled Robots

- Any microcontroller allowed
- Maximum dimensions:
  - 3kg: Max. of **20cm × 20cm**
  - No height limit
- Maximum weight: 3kg
- Any fixed sensors must be **covered with tape**

Section 3: Robot Restrictions

- **No jamming devices** targeting opponent's sensors
- **No damaging parts** for arena or opponent robot
- **No devices to throw liquids, powders, or gases**
  No sticky substances to increase traction
- **No devices to increase downforce** (vacuum or magnets)
- **Elementary Division:** BLADES are strictly **prohibited**

Section 4: Robot Placement

- Robots placed on **opposite sides**, facing outward
- Front of robot must touch the **white border**
- Robots start only upon **track monitor's instructions**

Section 5: Arena Specifications

- Arena: **4' × 4**
- **Material**: ~1.25 cm thick plywood or suitable nonmagnetic material
- Elevated by ~2.54 cm with support blocks (at least 1 cm from top edge)

Figure 1 : The Robo Push Competition Field
Adapted from Summer Robot Games 2025

Section 6: Game Rules

- Brackets determined based on number of teams:
  - <16 teams: 2 brackets
  - 16–32 teams: 4 brackets
  - 33 teams: 8 brackets

- **Robot Quarantine:** Teams must submit robots for inspection before competition
- Rules will be read by facilitator before each bracket
- Preliminary rounds:
  - Teams play against all other teams in their bracket
  - Brackets assigned randomly; results posted on Makerverse page
- Each match will consist of **one (1) round lasting three (3) minutes**. The robot that **pushes its opponent outside the arena within the allotted time** will be declared the **winner of the match**.

During Match

- Robot parts falling **inside arena**: match continues
- Robot parts falling **outside arena** (except screws/nuts): point to opponent
- **Repairs allowed** after each round: max. 1 minute, **no reprogramming**

- Battery replacement allowed after elimination rounds

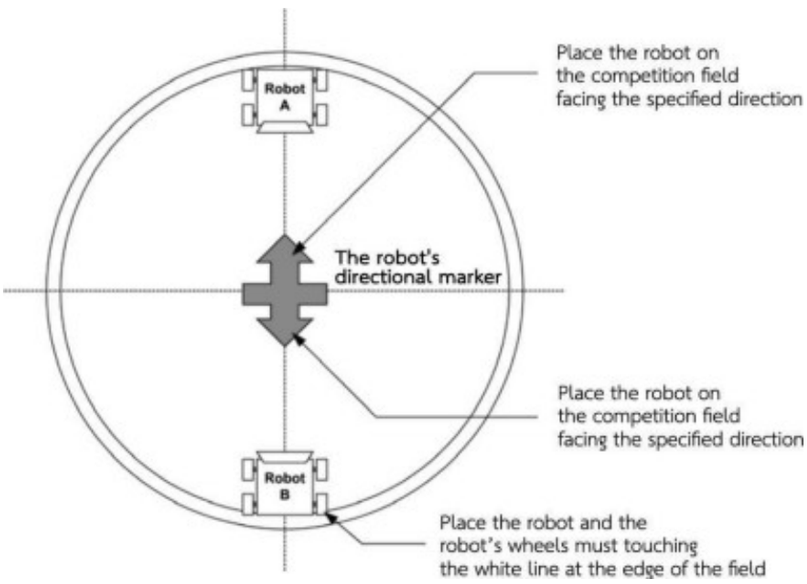

Figure 2: The Robots Placement Before the Start of the Match
Adapted from WRG 2026 PHL ONLINE Game Rules

Section 7: Tournament Structure

<16 teams:

- Two brackets, play all teams in bracket
- Top 4 teams advance to finals

16–32 teams:

- Four brackets, play all teams in bracket
- Top 2 from each bracket advance to finals

>33 teams:

- Eight brackets, play all teams in bracket
- Top 1 from each bracket advances to finals

Tie-breaker:

- Played in case of a tie after elimination rounds

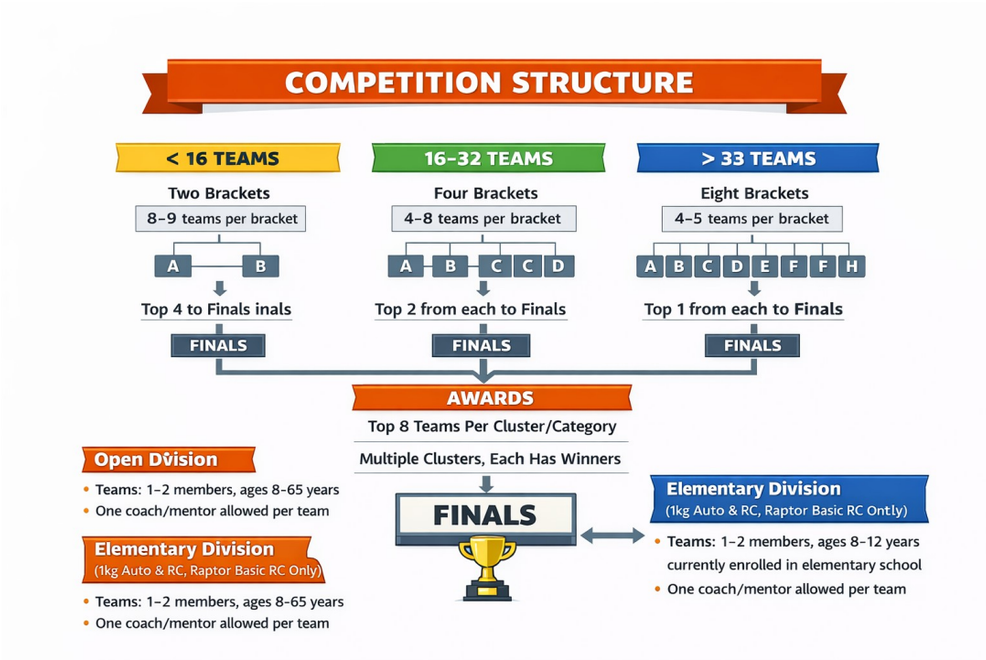

Figure 3: Competition Structure

Section 9: Judging and Conduct

- **Quarantine completion** is required to compete
- Teams failing to show after quarantine = **loss by default**
- Unsportsmanlike behavior may result in **disqualification**

Section 10: Final Authority

Note: *The referees are responsible for determining violations and enforcing penalties in accordance with the competition rules. All decisions made by the referees shall be final and binding.*

MakerBall

"Shoot for the Win"

Objective: The "Shoot The Ball" robot competition challenges participants to program a robot to follow a designated path while carrying a basket. The objective is to shoot as many balls as possible into the basket within **one (1) minute** while the robot is in motion.

Reference: World Robot Games Philippines. (2026). *World Robot Games 2026-PHL -Online GameRules and Maps* [PDF]. https://drive.google.com/drive/folders/1c5R7VDAZjqRO57JxGOiHQyiGtIo910ZG

Section 1: Participant Eligibility

1.1 Each team consists of **one (1) participant only**.
1.2 **Competition Categories**:

- Grades 1–3
- Grades 4–8

1.3 Participants may compete in **only one team and one division**.

Section 2: Competition Requirements

2.1 This competition is conducted **onsite at the Makerverse venue**.
2.2 Participants must bring:

- Fully assembled robot ready for play
- Measuring tools (ruler or tape measure)
- Stopwatch or timer (optional for participant use)
- Ping-pong balls for scoring missions
- Fully charged batteries + spare batteries
- Battery charger
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2.3 **Inspection:**

- Judges will inspect robots for size, weight, wheels, and ball container before each run
- Any modifications after inspection must be approved by judges

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Section 3: Playing Field Specifications

3.1 Field dimensions: **8’ × 10’**
3.2 Surface: Flat, made of **PVC, vinyl, or equivalent**
3.3 Black line width: ~25 mm
3.4 Includes designated area for **bouncing ping-pong balls**
3.5 Field must be placed on a **stable table**, height **no lower than participant’s waist**

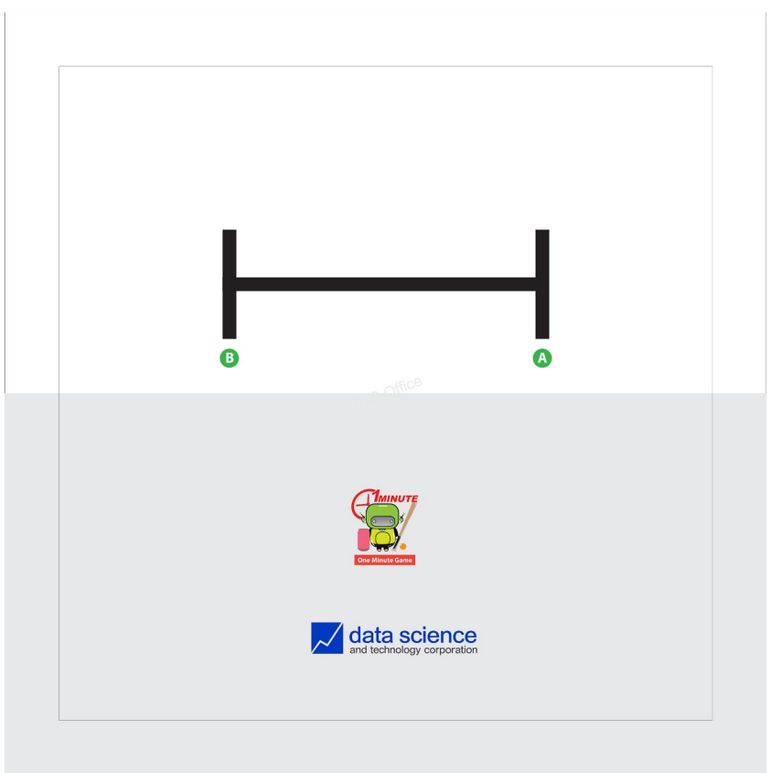

Figure 2: The Makerball Competition Field
Adapted from Data Science and Technology Corporation

Section 4: Robot Requirements

4.1 Maximum size: **20 × 20 cm**, no height limit, must fit inside a box for verification
4.2 Robot must use **wheels only**
4.3 Robot must move **autonomously**
4.4 No weight limit
4.5 All parts must be **securely attached**
4.6 Detached parts during run must **not be touched** until the run ends

Section 5: Ball Container Requirements

5.1 The robot must be equipped with a ball container:

- Maximum diameter: 15 cm
- Maximum height: 10 cm

5.2 The container must be designed to securely receive and retain the balls during operation
5.3 No materials or padding may be added inside the container
5.4 If a ball bounces out of the container:

- The ball does not count as a score
- The participant may retrieve and re-shoot the ball

5.5 Participants may replace the container if it becomes full
5.6 Previously collected balls must be kept separate for counting

Section 6: Controller and Components

6.1 Any microcontroller allowed
6.2 Motor & Sensor Limits:

- **Kinder-PreJunior / Junior Division:**
  -Maximum of **2 motors**
  -Maximum of **4 sensors**

6.3 Power supply & battery: **No limitation**

6.4 Ping-pong ball specifications:

- Diameter: 40mm

- Any color allowed
- Participants must bring sufficient balls

6.5 Ball container:

- Max diameter: 15 cm, max height: 10 cm
- Must securely hold balls without internal padding
- Balls bouncing out **do not count**
- Containers may be replaced if full
- Previously collected balls must be **kept separately**

Section 7: Prohibited Equipment

7.1 Any equipment that may **damage the playing field** is prohibited

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Section 8: Competition Format

8.1 Each participant gets **two (2)** official attempts
8.2 Teams are given **five (5) minutes** before the game for calibration
8.3 Marathon-style scoring: **highest valid run score holds "Best Score" title until surpassed**
8.4 Best valid score and time used for ranking
8.5 Ranking based on **highest score**, then total time including penalties
8.6 Tie-breakers: Faster time, then previous rounds' results

Section 9: Game Procedure and Mission

Mission:

- Robot moves along the line **Point A → Point B → Point A**
- Backward movement not allowed
- Minimum speed: **30 cm/s**
- Travel between points must not exceed **5 seconds**
- Participant stands **20 cm away** to shoot ping-pong balls:

    - Ball must **bounce once in designated area** then fall into container
    - Missed balls may be **re-shot**
- Total run time: **1 minute**

Game Start:

- Robot placed at Point A, facing any direction
- Timer started manually by participant
- Only participant may start robot and timer

During Run:

- Robot must follow the line autonomously
- Judges monitor progress and verify rules compliance

Scoring:

- Round trip (A → B → A) = **1 point**
- Each ball collected in container = **1 point**
- Final score = <round trips> + <balls collected>

End of Run:

- 1 minute expires
- Robot leaves the line completely
- Participant requests to stop
- Lack of progress >10 seconds

Section 10: Fouls and Unsportsmanlike Conduct

10.1 Disqualification for:

- Insulting or disrespectful behavior
- Interfering with other participants
- Entering another participant's area
- Bringing unauthorized items or causing disruptions

Section 11: Robot Identification

11.1 Each robot must display:

- Team name
- Participant name or ID

11.2 Must be visible to judges during all runs

Section 12: Judge Authority

12.1 Judges have authority over:

- Inspection of robot and playing field
  Run validity and scoring

12.2 Decision hierarchy:

1. Head Judge
2. Live observation review
3. Majority vote of judges

Note: *The referees are responsible for determining violations and enforcing penalties in accordance with the competition rules. All decisions made by the referees shall be final and binding.*

Track Mania

"Tip or Trip!"

Objective: The robot must reach the end of the playing track without falling or touching the floor.

Reference: Pilipinas RoboSports Tournament. (2026, February 5). *Pilipinas RoboSports Tournament version 8 rulebook* (Version 8) [PDF]. https://drive.google.com/drive/folders/1SwscBqoWkHreAloQ6HFQ4x7IYCx-Endd

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Section 1: Participants Eligibility

Open Division (RC)

1.1 Ages 8–65, 1–2 team members, 1 coach allowed.
1.2 Coaches can assist but cannot touch the robot during official runs.

Section 2: Robot Requirements

Remote-Controlled Robots

- Any programmable RC robot.
- Maximum dimension: **24 × 24 × 24 cm**.
- Sensors are **not allowed**.
- A remote controller must directly control the robot's movement.
- Non-detachable external sensors must be covered with electrical or masking tape.
- Use of built-in sensors (e.g., gyro, accelerometer) is **prohibited**.

Section 3: Playing Track

- Field Dimension: 6' x 10'

- Curves and angles may appear throughout the track.
- Track pieces connected with duct tape for stability.
- Start and End positions finalized with coaches before the competition.
- The Elementary Division may have **two(2) stations**; rankings based on station completion and speed.

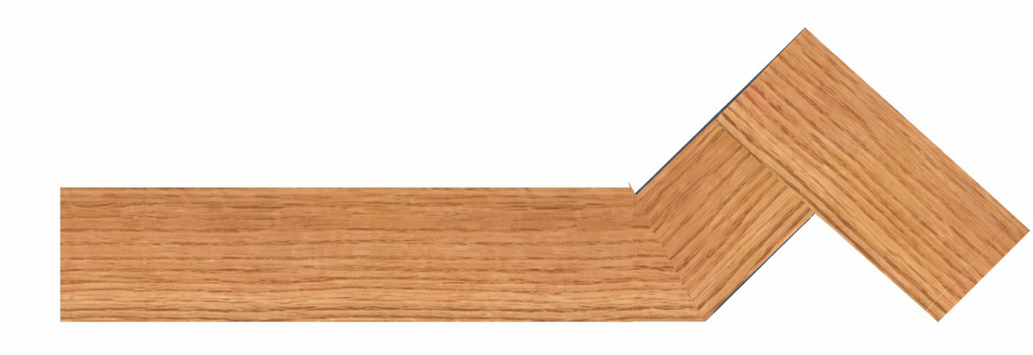

Figure 1 : The Robo Rush/Track Mani Game Track
Adapted from PILIPINAS ROBOSPORTS TOURNAMENT version 8

Note: *The official Track will be released before …*

Section 4: Game Procedure – Remote-Controlled

1. **Objective:** Reach the finish line under remote control without falling.
2. **Trials: one (1) trial** per official run; **2 official runs** allowed. (*suggestion*)
3. **Practice: one (1) minute** of practice allowed per run on official track.
4. **Timing:** Referee records time; fastest run used for ranking.
5. **Failure:** Robot falls or does not finish, run marked as “X”.
6. **Duration**: During the official run, participants will be provided up to **two (2) minutes** to reach the finish line.
7. Teams are given **five (5) minutes** before the game for calibration

Figure 3: Competition Structure

Section 5: Scoring

1. **Ranking:** Based on fastest successful official run.

Section 6: Conduct & Fouls

1. Unsportsmanlike behavior or interference leads to **disqualification**.
2. Teams must follow instructions during practice and official runs.

3. Participants may **not step on the playing track**.
4. Failure to comply will result in team disqualification.

Section 7: Judge Authority

1. Judges monitor: Robot, track, and official timing.
2. Decisions hierarchy: Head Judge → Observation → Majority vote.

Note: *The referees are responsible for determining violations and enforcing penalties in accordance with the competition rules. All decisions made by the referees shall be final and binding.*

SoccerBot

"Kick, Score, Win the Game"

Objective: SoccerBot is for remote-controlled or autonomous robots, each weighing no more than 1 kg, to push or kick the ball into the opponent team's goal.

Reference: Pilipinas RoboSports Tournament. (2026, February 5). *Pilipinas RoboSports Tournament version 8 rulebook* (Version 8) [PDF]. https://drive.google.com/drive/folders/1SwscBqoWkHreAloQ6HFQ4x7IYCx-Endd

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Section 1: Participants

Open Division (RC):

1.1 **Team Composition:** Teams shall be composed of **three (3) members**, aged **8 to 65 years old**.
1.2 **Coach:** Only 1 coach allowed per team.

Section 2: Robot Requirements

1.1 **Microcontroller:** Any, programmable or non-programmable.
1.2 **Robot Dimensions:** Maximum **20 × 20 cm**, no height limit.
1.3 **Robot Weight:** Maximum **1 kg**.
1.4 **Design Requirements:**

- Kicking or ball-pushing mechanisms must be **covered with tape** to prevent damage to the arena or ball.

Section 3: Robot Restrictions

Robots must **not include**:

1. Jamming devices that affect opponent robot connectivity.
2. Parts that can break or damage the arena or goal.
3. Parts intended to damage opponent robots or operators. Normal pushing allowed.
4. Devices storing liquids, powder, gases, or other substances for throwing.

5. Sticky substances for traction improvement.
6. Devices to increase downforce (vacuum pumps or magnets).

Section 4: Arena Specifications

- Size: Approximately **6’ x 12’**
- Field surface: Printed tarpaulin or mat.
- Goals: Standard size to match arena width.
- Boundaries clearly marked.

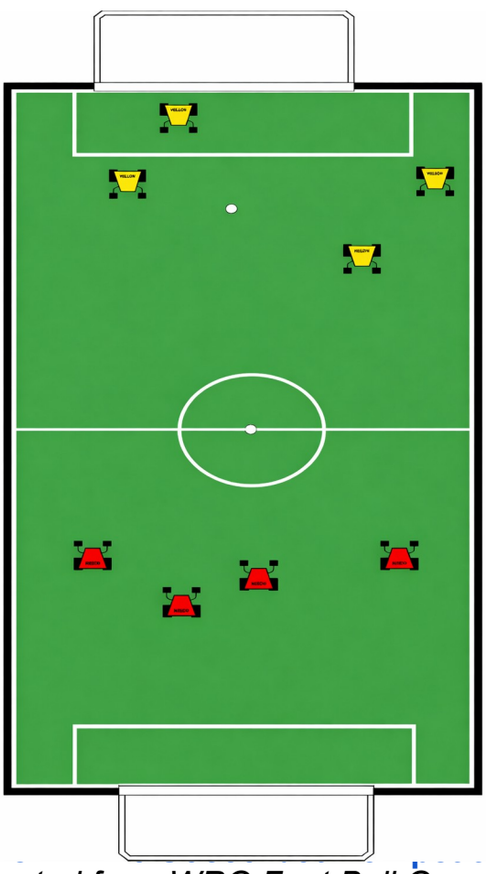

Fig Field

Adapted from WRG Foot Ball Game Rules

Section 5: Game Procedure

1. **Match Type: Single elimination**; group stages may be applied if necessary.
2. **Team Assignment:**
    - Teams assigned via **digital random sorter**.
    - “BY Teams” or bytes determined through **draw lots** if required.
3. **Battery Replacement:** Allowed **after each match**.
4. **Match Time: 3–5 minutes** per round, determined by organizers.
5. **Game Conduct:**
    - Teams may only touch their robots for control.
    - No interference with opponent robots outside gameplay.
6. **Ball Handling:** Robots may **push or kick the ball** using built-in mechanisms; throwing, lifting, or dropping is prohibited.

Section 6: Scoring & Awards

1. **Ranking Basis:** Number of wins in elimination or group rounds
2. **Awards:**
    - Top 8 teams per category/division.
    - Certificates and gold medals.

Precision Striker Award: Presented to participants whose robot demonstrates exceptional accuracy and control in shooting, consistently hitting the target with skillful precision.

Section 7: Conduct & Fouls

1. Unsportsmanlike behavior or interference leads to **disqualification**.
2. Teams must follow official instructions during practice and official matches.
3. Intentional damage to robots, arena, or opponent leads to **immediate disqualification**.

Section 8: Judge Authority

1. Judges monitor gameplay, arena condition, and scoring.
2. Decision hierarchy: **Head Judge → Observation → Majority vote**.

Note: *The referees are responsible for determining violations and enforcing penalties in accordance with the competition rules. All decisions made by the referees shall be final and binding.*

Makerverse Robotics Innovation

"Innovate, Build, Inspire"

Objective: Teams design, build, and showcase a theme-based robotic project that demonstrates creativity, problem-solving, and functional autonomous or semi-autonomous behavior.

Reference: World Robot Games Philippines. (2026). *World Robot Games 2026-PHL Online GameRules and Maps* [PDF]. Summer Robot games 2025 https://drive.google.com/drive/folders/1c5R7VDAZjqRO57JxGOiHQyiGtIo910ZG

Competition Theme: "Health and Disaster Response"

3 sub-projects:

- Firefighting System
- Automatic Waste Bin System
- Smart Clothes Wire

Section 1: Participants

1.1 **Team Composition:** 1–3 student members per team.
1.2 **Team Limit:** Each participant may compete in only one team.
1.3 **Project Submission:** Each team submits one project to demonstrate live at the venue.
1.4 **Competition Categories**:

- Grade 4-8
- Grade 9-12
- Open Category

Section 2: Robot Requirements

2.1 Grade 4-8

- **Motor Limit**: Max. of **four (4)**
- **Sensor Limit:** Max of **four (4)**
- Any Microcontroller

2.3 Grade 9-12

- **Motors:** No Limit
- **Sensors:** No Limit
- Multiple control boards allowed
- Power supply unrestricted
- All construction/programming must be by students
- Only approved components

2.4 Open Division

- **Motors:** No Limit
- **Sensors:** No Limit
- Multiple control boards allowed
- Power supply unrestricted
- All construction/programming must be by students
- Only approved components

Section 3: Equipment and Preparation

2.1 Firefighting Robotics System
Objective: Create a Smart Fire-Response System

Core Components

- **Microcontroller**: Nano
- **Motors**: Servo, Water Pump
- **Sensors**: IR Flame Sensor, MQ-2 Smoke Sensor

Note: *Quantities must meet design and technical requirements, ensuring all components fit the robot and comply with competition rules.*

General Materials (optional)

- **Wiring:** Jumper wires, alligator clips, breadboard, connectors
- **Power:** Battery pack, ON/OFF switch, voltage regulator (if needed)
- **Structure:** Chassis/frame using cardboard/cartoon, plywood, acrylic, or 3D-printed parts; mounting brackets, screws, zip ties
- **Fire Suppression:** Tubing, mini nozzle/sprinkler, water reservoir
- **Tools & Assembly:** Soldering iron, pliers, screwdrivers, hot glue, wire stripper

Note: *Participants may combine materials for stability, durability, and mobility while keeping the robot lightweight.*

2.2 Automatic Waste Bin Robotics System

Objective: Automatically sorts waste for recycling

Core Components

- **Microcontroller**: Arduino / ESP32
- **Sensors**: IR Sensor, Ultrasonic Sensor
- **Actuator**: Servo Motor

Note: *Quantities must meet design and technical requirements, ensuring all components fit the system and comply with competition rules.*

General Materials (optional)

- **Power**: Battery pack or adapter
- **Wiring & Connections:** Jumper wires, connectors
- **Structure**: Chassis/frame using cardboard, plywood, acrylic, or 3D-printed parts; mounting brackets, screws, zip ties
- **Tools & Assembly**: Soldering iron, pliers, screwdrivers, hot glue, wire stripper

Note: *Participants may combine materials for stability, durability, and mobility while keeping the system lightweight.*

2.3 Smart Clothes Wire

Objective: Build a rain-sensing robot that automatically folds a clothes wire to protect clothes

Core Components

- **Microcontroller**: Arduino / ESP32

- **Sensors**: Rain Sensor, IR Sensor (optional for position detection)
- **Motors**: Servo Motor or DC Motor with gears (to fold/unfold clothes wire)
- **Actuators / Mechanical Parts:** Pulley, string, or linkage for wire movement
- Clothes **Wire** / **Rod**

Note: Q*uantities must meet design and technical requirements, ensuring all components fit the system and comply with competition rules.*

General Materials (optional)

- **Power**: Battery pack or adapter
- **Wiring & Connections:** Jumper wires, connectors
- **Structure**: Frame using cardboard, plywood, acrylic, or 3D-printed parts; mounting brackets, screws, zip ties
- **Tools & Assembly**: Screwdrivers, pliers, soldering iron, hot glue, wire stripper

Note: *Participants may combine materials for stability, durability, and mobility while keeping the system lightweight and safe.*

Section 4: Project Display Setup

- Each participating project is allocated **1 folding table**.
- Participants are responsible for arranging their project neatly on their table.
- Ensure all project items are secure and stable during the competition.
- Participants should **stand by their table** during judging

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Section 5: Live Demonstration Guidelines

- **Duration:** 5 minutes per team.
- **Requirements:**
    1. Introducing the team members.
    2. Explain the project concept and robot functionality.

3. Demonstrate robot physical operation and intelligent behavior live.
4. Explain hardware, programming, and design decisions.

- **Judges' Questions:** Teams may be asked to answer questions after demonstration.
- **Robots must operate live on-site:** no pre-recorded videos are allowed.

Criterion	Weight	Description
Theme Alignment & Project Purpose	40%	Clear objectives, demonstration of intelligence, alignment with theme
Robot/System Design & Technical Implementation	30%	Functionality, reliability, mechanical/electronic design, programming complexity, integration
Presentation Quality	30%	Clarity, visuals, communication, understanding of design decisions

Judges evaluate the students' understanding and execution of their project. Originality alone is not the main criterion.

Section 6: Conduct & Safety

- Teams must follow all competition rules and judges' instructions.
- Unsafe behavior, interference with other teams, or unsportsmanlike conduct may result in disqualification or score deduction.

Section 7: Awards

- **Winners** in each division will be presented with Makerverse trophies, medals, and certificates in recognition of their outstanding performance.
- **Special awards** may include:

- Excellence in Innovation Award
- Future Innovator Award
- Best Team Award

Section 8: Judge Authority

- Judges have full authority over **scoring, evaluation, and ruling disputes**.
- The **Head Judge's decision is final and non-appealable**.

Note: *Teams are encouraged to implement innovative designs and solutions. However, all robots must comply with the competition rules. Any violations, unsafe designs, or non-compliant modifications will be subject to penalties at the referees' discretion.*

1st DIPOLOG ROBOTICS FESTIVAL AND COMPETITION 2026

OFFICIAL REFEREE MANUAL

GENERAL RULES FOR ALL GAMES

1. The referee's decision is FINAL.
2. No coaching during match/run.
3. 5-minute repair time allowed.
4. Unsportsmanlike conduct = Warning.
5. 2 Warnings = Disqualification.

START PROCEDURE

1. Call participants.
2. Confirm readiness.
3. Count down:
   “3… 2… 1… START!”
4. Activate timer.
5. Monitor violations.

STOP CONDITIONS

- Time limit reached
- Robot leaves arena
- Robot immobilized
- Manual interference

PENALTIES

Minor Violation = +5 seconds
Major Violation = +10 seconds
Unsafe Robot = Immediate stop

LINE DASH

Penalties may be applied during the run depending on the severity of the violation, as outlined in the table below.

Violation Type	Description	Penalty
Minor Violation	✓ Slightly leaving the line ✓ Small deviation from path	+ 5 seconds to run time
Major Violation	✓ Repeatedly leaving	+ 10 seconds to run time

	the track ✓ Ignoring track boundaries	
Unsafe Robot	✓ Sparking motors ✓ Parts falling off	Run stopped immediately; robot removed from track

Note: *All penalties will be assessed at the discretion of the referees, and their decisions shall be final.*

ROBO PUSH

Notes: *Penalties will be applied during the run based on the severity of any rule violations, as shown in the table below.*

Violation Type	Description	Penalty	Referee Notes
Minor Violation	✓ Slightly moving before the referee's signal ✓ Small arena infractions (touching border, minor accidental contact with opponent without control)	+5 seconds added to match time or warning	Apply once per incident; monitor for repeat offenses

Major Violation	✓ Repeatedly moving before start ✓ Intentionally damaging arena or opponent robot ✓ Use of prohibited parts (blades, sticky substances, jamming devices)	+10 seconds added to match time or possible disqualification	Enforce consistently; referee discretion required
Unsafe Robot	✓ Sparking motors, exposed wires, or loose parts ✓ Using unsafe mechanisms (blades, projectiles)	Immediate stop of match; robot removed from arena	Stop match immediately; prioritize safety above all

Notes: *If a robot completes the push in **50 seconds**, but commits a minor violation:*

- Final recorded time = **50 + 5 = 55 seconds**

Maker Ball

Notes: *Penalties will be applied during the run based on the severity of any rule violations, as shown in the table below.*

Violation Type	Description	Penalty
Minor	✓ Slight line deviation ✓ Minor disruption	Deduct **1 ball** from total score
Major	✓ Repeated line leaving ✓ Prohibited equipment	Deduct **2 balls** from total score
Unsafe Robot	✓ Detached or loose parts ✓ Hazardous components	Immediate stop; robot removed

Note: *Judges have full authority to determine violations and enforce penalties. All decisions are final and binding.*

Track Mania

Notes: *Penalties will be applied during the run based on the severity of any rule violations, as shown in the table below.*

Violation Type	Description	Penalty
Minor	✓ Slight deviation from track without falling	+5 sec. added to run time or warning
Major	✓ Participants interferes with the robot during the official run	+10 sec. added to run time
Unsafe Robot/Fall	✓Robot falls off track	Run stopped immediately; robot removed from track; run marked as “X”

Note: *Judges have full authority to determine violations and enforce penalties. All decisions are final and binding.*

Soccerbot

Note: Penalties may be applied during the match for violations of the rules. Judges monitor robots, gameplay, and participant actions. Time penalties or match rulings will be applied based on the severity of the infraction.

Violation Type	Description	Penalty
Minor	✓ Temporary robot malfunction (robot stops or freezes briefly but is safe)	Deduct 1 goal/ball from team score
Major	✓Stocking or holding opponent robot ≥8 seconds	Deduct 3 goals/balls from team score
Unsafe Robot	✓ Detached or loose parts ✓Hazardous components affecting players or arena	Immediate stop of match; robot removed from play

Innovation

Note: *Penalties may be applied for rule violations, safety issues, or non-compliance. All deductions are at the judges' discretion and are final.*

Violation Type	Description	Penalty
Minor	✓ Minor rule deviations ✓ Slight safety or design issues	Deduct **1 point** from total score
Major	✓ Significant rule violations ✓ Exceeding the allotted 5-minute presentation time	Deduct **3 points** from total score
Unsafe/Disqualified	✓ Hazardous designs or safety risk	Immediate disqualification

Note: *Minor and major penalties reduce the participant's overall Innovation score. All decisions are at the discretion of the judges and are final.*